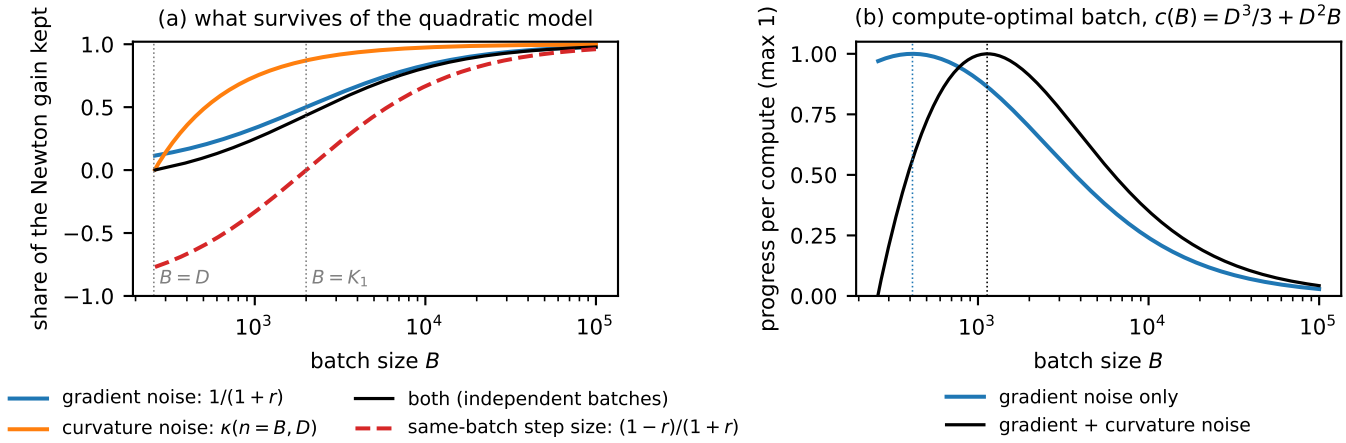
Illustration: $D = 256$, $K_1 = 2000$ examples, $N \to \infty$, exact Newton direction
(a) what survives of the quadratic model
share of the Newton gain kept
1.0
0.5
0.0
−0.5
−1.0
$B = D$
$B = K_1$
10^3
10^4
10^5
batch size B
gradient noise: $1/(1+r)$
curvature noise: $\kappa(n = B, D)$
both (independent batches)
same-batch step size: $(1-r)/(1+r)$
(b) compute-optimal batch, $c(B) = D^3/3 + D^2B$
progress per compute (max 1)
1.00
0.75
0.50
0.25
0.00
10^3
10^4
10^5
batch size B
gradient noise only
gradient + curvature noise